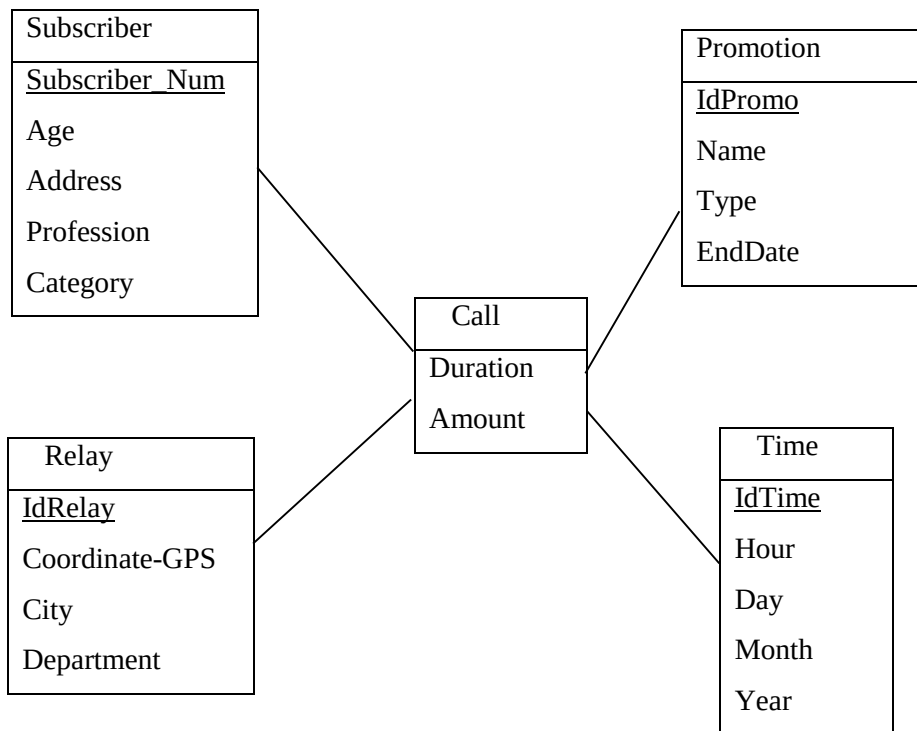


Tuto #5

Exercise 01:

A mobile phone operator has designed a Data Warehouse (DW) for its Business Intelligence System. The conceptual Schema of this DW is as follows:



Remarks :

- Two types of promotions are possible: “prepaid” and postpaid”
- The table “Call” contains 100,000,000 records.

QUESTIONS :

1. Queries and Optimization

- Give the logical model of this data warehouse.
- Let consider the following queries :
 - **Q1** : The total amount of calls made by subscribers aged over 20 years.
 - **Q2** : The total amount of calls made by subscribers aged over 20 for type promotions of type “Prepaid”.
 - **Q3**: The average amount of calls made by subscribers aged over 20 passing through relays located in the city of Algiers.
 - What is the algebraic tree of each query ?
 - What is the global algebraic tree grouping the three queries at same time?
 - Give the algebraic expressions of all candidate materialized views to optimize these queries.
 - Which is the most interesting materialized view, justify your answer ?
 - Give the SQL query that creates this view.

6. Re-write the queries Q1, Q2 and Q3 regarding this view.

2. Fragmentation

1. Propose a fragmentation of the table “Subscriber” to optimize the query Q1. Give the algebraic expressions.
2. Propose fragmentation of the table “Calls”. Give the algebraic expressions.
3. Give the scripts of fragmentation for the tables “Subscriber” and “Call”.
4. Rewrite the queries Q1 and Q3 on the fragmented schema.

3. Bitmap Index

1. Propose a **join bitmap index** to optimize the query Q2.
2. Calculate the size of this index in KB
3. How will this index be used to answer query Q2 ?